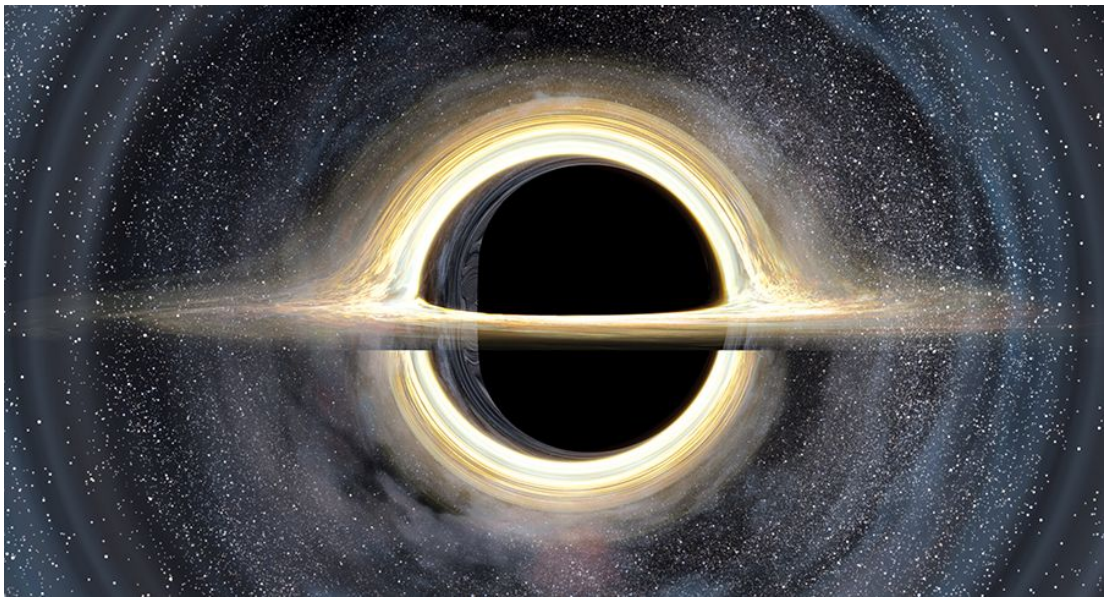


Fig. 9.9 Gargantua with the infinitesimally thin paint-swatch disk (Fig. 9.7) replaced by a more realistic, infinitesimally thin accretion disk. *[From Eugénie von Tunzelmann's artistic team at Double Negative.]*

The artistic team at Double Negative then gave the disk the texture and surface relief that we expect a real, anemic accretion disk to have, puffing it up a bit in a manner that varied from place to place. They made the disk hotter (brighter) near Gargantua and cooler (dimmer) at larger distances. They made it thicker at larger distances because it is Gargantua's tidal gravity that squeezes the disk into the equatorial plane, and tidal gravity is much weaker farther from the black hole. They added the background galaxy: many layers of artwork (dust, nebulae, stars). And they added lens flare—the haze and glare and streaks of light that would arise from scattering of the disk's bright light in a camera lens. The results were the wonderful and compelling images in the movie (Figures 9.10 and 9.11).